The Economics of Bicycles for the Mind

Cognitive tools, human judgment, and inequality

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Agrawal, Gans & Goldfarb (2025) · NBER WP 34034

github.com/gsaco/ai-02-agrawal

The paper and the agent's problem

Question and single mechanism

Question. When a cognitive tool makes implementation easier, how do effort, productivity, returns to human abilities, and inequality change?

Mechanism. The tool is a substitute for implementation effort at the margin; implementation skill s, payoff judgment α, and opportunity judgment $\gamma(t)$ determine who can exploit it.

Agent's problem, conditional on finding an opportunity

Choose effort e_t to maximize expected improvement value net of cost:

$$\boxed{e_t^*(\theta) \in \arg\max_{e_t \geq 0} \{p(se_t;\theta)\alpha\Delta - c(e_t;\theta)\}.}$$

Interior FOC: $p'(se_t^*;\theta)s\alpha\Delta = c'(e_t^*;\theta)$.

Choice and constraint: $e_t \geq 0$.

Technology: $p \in [0,1]$ is increasing and weakly concave.

Cost: c is increasing and weakly convex.

Tool: raises p and/or lowers c, while lowering p'/c'.

Proposition 3: the conditional variance result

Specialization and conditions

$p(se;\theta) = \sqrt{se+\theta}$, $c(e) = e$; $\Gamma = \gamma_0/(1-\delta\gamma)$ and $\delta\gamma < 1$.

$\alpha, \gamma_0, \gamma, s$ are independent, have positive support, and satisfy $\mu_i > 3\sigma_i$.

Common interior range: $\theta < \alpha^2\Delta^2 s^2/4$ for every worker under comparison.

Corrected exact statement

$$V(\theta) = \Gamma\left(\frac{\alpha^2\Delta^2 s}{4} + \frac{\theta}{s}\right).$$

$$\frac{d\mathbb{E}[V]}{d\theta} = \mathbb{E}[\Gamma]\mathbb{E}[1/s] > 0.$$

The U-shape concerns cross-sectional $\mathrm{Var}[V(\theta)]$ as continuous tool quality θ changes.

If

$$\frac{\mathbb{E}[\Gamma^2]}{\mathbb{E}[\Gamma]^2} < \mu_s\mathbb{E}[1/s], \qquad \mathrm{Var}(\Gamma/s) > 0,$$

then $\mathrm{Var}[V(\theta)]$ is strictly convex, falls for $\theta < \theta^*$, and rises for $\theta > \theta^*$, where $\theta^* = -a_0/[2\,\mathrm{Var}(\Gamma/s)] > 0$.

Endpoint correction and intuition

Equation (32)'s positive slope at $\theta = 1$ also requires $\theta^* < 1$. **Intuition:** inverse skill bias equalizes first; heterogeneous opportunity judgment can dominate later.